Mathematics in the Modern World

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Types of Propositions

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- Propositions can be divided into 3 types based on their logical features:
  - **Tautology**
    - A compound proposition is a tautology or logically true if it is true regardless of the truth-values of its components.
  - **Self-contradiction**
    - A compound proposition is said to be self-contradictory or logically false if it is false regardless of the truth-values of its components.
  - **Contingency**
    - A compound proposition is said to be contingent if its truth-values depend on the truth-values of its components.

Propositional Equivalences

DEFINITION 1

- A compound proposition that is always true, no matter what the truth values of the propositions that occurs in it, is called a ***tautology***.
- A compound proposition that is always false is called a ***contradiction***.
- A compound proposition that is neither a tautology or a contradiction is called a ***contingency***.

Examples of a Tautology and a Contradiction			
p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

Proposition type	Column under main operator
Tautology	All true
Contradictory	All false
Contingent	At least one true, at least one false

Relations between Propositions

- Two propositions are **(logically) equivalent** if they have the same truth-values regardless of the truth-values of the components.

- Two propositions are **(logically) contradictory** if they have opposite truth-values regardless of the truth-values of their components.

Propositional Equivalences

Logical Equivalences

DEFINITION 2

- The compound propositions p and q are called ***logically equivalent*** if $p \leftrightarrow q$ **is a tautology.**
- The notation $p \equiv q$ denotes that p **and** q **are logically equivalent.**

Propositional Equivalences

Logical Equivalences

- Compound propositions that have the same truth values in all possible cases are called **logically equivalent**.

Example: Show that $\neg p \vee q$ and $p \rightarrow q$ are logically equivalent.

Truth Tables for $\neg p \vee q$ and $p \rightarrow q$.				
p	q	$\neg p$	$\neg p \vee q$	$p \rightarrow q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Propositional Equivalences

- In general,

  2^n rows are required if a compound proposition involves n propositional variables in order to get the combination of all truth values.

Tautological Equivalences

- Two statement forms are called *logically equivalent* if, and only if, they have identical truth values for each possible substitution of statements for their statement variables.
- The logical equivalence of statement forms P and Q is denoted by writing $P \Leftrightarrow Q$ or $P \equiv Q$.
- Truth table for $(A \vee B) \vee C \Leftrightarrow A \vee (B \vee C)$

A	B	C	A V B	B V C	(A V B) V C	A V (B V C)
T	T	T	T	T	T	T
T	T	F	T	T	T	T
T	F	T	T	T	T	T
T	F	F	T	F	T	T
F	T	T	T	T	T	T
F	T	F	T	T	T	T
F	F	T	F	T	T	T
F	F	F	F	F	F	F

Some Common Equivalences

- The equivalences are listed in pairs, hence they are called duals of each other.
- One equivalence can be obtained from another by replacing V with Λ and 0 with 1 or vice versa.

Commutative	$A \vee B \Leftrightarrow B \vee A$	$A \wedge B \Leftrightarrow B \wedge A$
Associative	$(A \vee B) \vee C \Leftrightarrow A \vee (B \vee C)$	$(A \wedge B) \wedge C \Leftrightarrow A \wedge (B \wedge C)$
Distributive	$A \vee (B \wedge C) \Leftrightarrow (A \vee B) \wedge (A \vee C)$	$A \wedge (B \vee C) \Leftrightarrow (A \wedge B) \vee (A \wedge C)$
Identity	$A \vee 0 \Leftrightarrow A$	$A \wedge 1 \Leftrightarrow A$
Complement	$A \vee A' \Leftrightarrow 1$	$A \wedge A' \Leftrightarrow 0$

- Prove the distributive property using truth tables.

DE MORGAN'S LAWS

- The negation of an and statement is logically equivalent to the or statement in which each component is negated.

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De Morgan's Laws

1. $(A \vee B)' \Leftrightarrow A' \wedge B'$
2. $(A \wedge B)' \Leftrightarrow A' \vee B'$

A	B	A′	B′	A V B	(A V B)′	A′ Λ B′
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

- Conditional Statements in programming use logical connectives with statements.

Propositional Equivalences

Constructing New Logical Equivalences

- **Example:** Show that $\neg(p \rightarrow q)$ and $p \wedge \neg q$ are logically equivalent.

  Solution:

$\neg(p \rightarrow q) \equiv \neg(\neg p \vee q)$ *by example*

$\equiv \neg(\neg p) \wedge \neg q$ *by the second De Morgan law*

$\equiv p \wedge \neg q$ *by the double negation law*

- ***Example:*** *Show that* $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

  Solution: To show that this statement is a tautology, we will use logical equivalences to demonstrate that it is logically equivalent to T.

$(p \wedge q) \rightarrow (p \vee q) \equiv \neg(p \wedge q) \vee (p \vee q)$ *by example*

$\equiv (\neg p \vee \neg q) \vee (p \vee q)$ *by the first De Morgan law*

$\equiv (\neg p \vee p) \vee (\neg q \vee q)$ *by the associative and communicative law for disjunction*

$\equiv T \vee T$

$\equiv T$

- ***Note:*** ***The above examples can also be done using truth tables.***

Directions: Determine whether each of the following propositions is a tautology, a contradiction, or a contingency.

1. $(p \rightarrow q) \wedge (q \rightarrow p) \leftrightarrow (p \leftrightarrow q)$

2. $\sim(p \rightarrow q) \wedge (p \wedge \sim q)$

3. $[\sim(p \rightarrow r) \rightarrow \sim q] \rightarrow [\sim(p \rightarrow r) \vee q]$

4. $[p \wedge (p \rightarrow q)] \rightarrow \sim q$

5. $(p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p)$